Learn Where It Pays: Certified-Safe, Headroom-Gated Learning for Workplace EV Charging on Radial Feeders

Research proposal v2 — revision after the empirical audit of the Residual-Dyna / SafeTD3 programme

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Abstract

The original proposal predicted that a differentiable residual-Dyna SafeTD3 agent would cut cost by ∼50% and converge 4–10× faster than baselines. After building the full pipeline on real ACN-Data (Caltech, JPL), the IEEE-33 feeder and a cited tariff, and after a deliberate audit (14 shortcomings found and fixed, three multi-seed held-out studies), the evidence says something different and, we argue, more publishable: (i) the *safety* machinery is solid and novel in its details — row-wise split-conformal margins that absorb the LinDistFlow→AC mismatch and an exact feasibility-restoration step give zero violations on every policy that passes through them, whereas raw greedy dispatch overloads the feeder by up to 0.31 p.u.; (ii) the *learning headroom* is small — on 20 real days with a demand charge and a stressed hub a hindsight oracle beats a one-parameter tuned EDF cap by only 5.2%, and a collaborator's four-arm study (4.8–12%) agrees; (iii) a safe *residual* SafeTD3 around the tuned EDF removes the late-training degradation but, after 60 episodes, stays 2–9% *worse* than the heuristic it corrects. We therefore re-position the work as **(A) certified-safe dispatch with finite-sample guarantees, (B) a headroom-gated learning protocol and benchmark, and (C) a validation-gated residual policy that is provably not deployed unless it helps**, with a recursive tree-decomposition of the safety layer as the scale-up path to larger feeders. Cross-site (Caltech→JPL) transfer and Caltech+JPL federated results, in stressed and unstressed regimes, are reported in §6. Every number below is tagged with where it came from.

1 Where we are, in one page

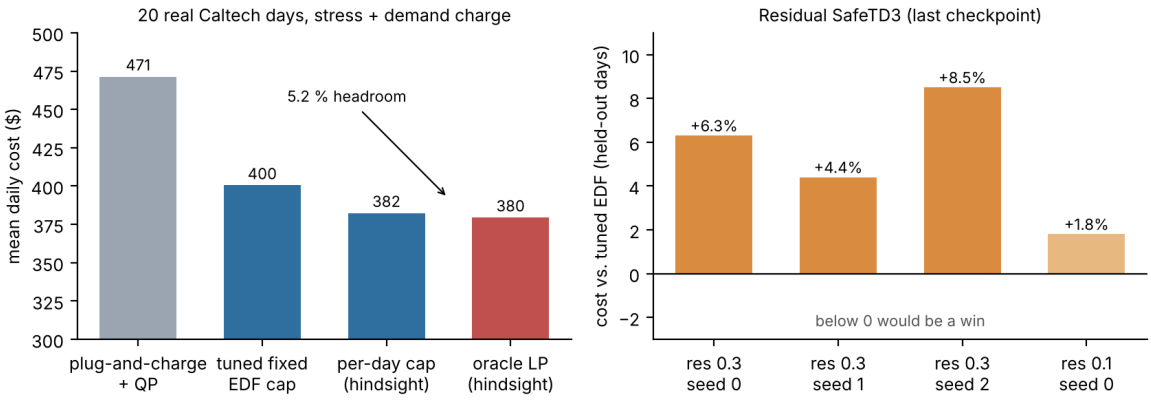


Figure 1: Left: costs on 20 real Caltech weekdays (stress + demand charge 1.5 $/kW-day), no learning involved. The oracle's whole advantage over a tuned fixed EDF cap is 5.2%; one scalar chosen per day in hindsight already reaches it. Right: residual SafeTD3 relative to the tuned EDF on held-out days (3 seeds at scale 0.3, one at 0.1): never below zero.

- **Works and is ours:** differentiable QP safety layer that is now fast and exact (row generation, float32-safe ADMM, implicit KKT gradients, feasibility restoration); row-wise conformal margins (fixed a policy

collapse caused by a global margin of 45.8 kVA against 73–128 kVA branch headroom); dense potential-based shaping reward; multi-day train/val/test protocol; oracle LP; demand-charge and stress scenarios; 68 unit tests.

- **Does not work (yet):** learning beyond a tuned EDF cap. **Does not exist in the data:** V2G profit (oracle gain $< 0.6\%$ under 10% round-trip loss, degradation and demand charges — with *assumed* battery parameters, because ACN has none).
- **Consequence:** the contribution must be the safety certificate, the benchmark/headroom methodology, and a gated residual learner — not "RL beats heuristics". This is a stronger paper than an unfalsifiable performance claim because every reviewer who runs a tuned EDF baseline would otherwise sink it.

2 Evidence register (what we know, with provenance)

ID	Finding	Evidence	Source
E1	Greedy/ToU dispatch overloads branches under stress (0.09–0.31 p.u.); every QP-wrapped policy has 0	`docs/study/headroom_*.json`, `M*`/`R*` JSONs	run
E2	On unstressed real days plug-and-charge is within 1–3% of the oracle on 14/16 cases	`headroom_oracle_vs_safe_uncontrolled.json`	run
E3	20 days, stress + demand: safe-uncontrolled 471.4; best fixed EDF cap 400.5; per-day cap 382.4; oracle 379.5	`headroom_multiday.json`	run
E4	Day-level cap forecast does not beat one fixed cap: ex-ante context 404.0, clairvoyant day energy 393.2 vs 400.5	`scripts/poc_cap_selection.py`	run
E5	Price-blind online peak shaving (OA) is worse than a fixed cap (471.7–496.4 vs 400.5)	`oa_peak` rule	run
E6	Plain SafeTD3 on held-out days: 313–443 (best/last), default config 651; late-training degradation in every setup	`M1,M2,R2`	run
E7	Safe residual SafeTD3 around tuned EDF: last-checkpoint $+6.3, +4.4, +8.5\%$ (3 seeds, scale 0.3), $+1.8\%$ (scale 0.1); noise, Optuna params, margin off: no change	`R1,R1s1,R1s2,R3,R4,R5,R6`	run
E8	Global conformal margin collapsed the policy; per-row margins fix it	`legacy_margin_collapse/`	run
E9	Four-arm study (ToU/CAISO-like × with/without V2G), normal and stress: headroom 4.8–12.0%, V2G oracle gain ≤ 0.9\$/day, naive V2G greedy +26 to +52% worse	collaborator DOC.md §13, `REPRODUCIBILITY.md`	**not re-run by me**
E10	Optuna "291.9" is a training-day validation cost; on the same day plain charging costs ∼279; on held-out days the same parameters give 363–376	`antigravity_optuna_best.json`, `R2`	run

Reading E9 carefully. (a) Its "CAISO" price is a calibrated *synthetic* duck curve, not CAISO data; we say "synthetic CAISO-like" until a real SP15 LMP CSV is loaded (§8). (b) Its V2G gain depends on assumed battery capacity, efficiency and degradation. (c) The claim "static cap 0.40 collapses under stress, hence RL is needed" is a *re-tuning artefact*: a cap tuned on the stressed training days is fine ($\leq$12% from the oracle), and our own E7 shows learning does not currently beat the re-tuned cap. The defensible version is: a cap needs re-tuning per regime and per site, which is exactly what the headroom-gated protocol automates (C3).

3 Decision register: keep, re-scope, discard

Component	Verdict	Reason / what changes
Differentiable QP safety layer, hard-first + row generation + restoration	**KEEP**	Core contribution; exact certificate (Prop. 2); 3–10× faster than the slack-ADMM path.
Row-wise split-conformal margins	**KEEP**	Finite-sample coverage (Prop. 1); fixes collapse; extend with adaptive conformal for shift (§5).
Dense potential-based shaping reward	**KEEP**	Policy-invariant (Ng et al. 1999); cured the sparse-reward failure (S1).

Component	Verdict	Reason / what changes
Residual world model ($F_{\text{base}} + N_\theta$)	**RE-SCOPE**	Accurate (voltage MAE $\sim 10^{-4}$ p.u.) but the model-based actor term gave no measured gain and is inconsistent with a demand charge ($\lambda_{mb} = 0$ in all demand arms). Keep as a safety-margin estimator; test as a policy-gradient aid only in an ablation.
SafeTD3 on the 108-d action, learned from scratch	**DISCARD** as headline	Never beat a tuned EDF; degrades late. Keep only as an ablation arm.
Residual SafeTD3 around tuned EDF	**RE-SCOPE**	Stable, safe, 2–9% above EDF today. Becomes the *gated* learner (C4): deployed only if a validation test proves improvement.
Greedy anchor (TD3+BC)	**DISCARD**	Hurt on held-out days (M1). Superseded by the residual base.
V2G arbitrage / V2G-based claims	**RE-SCOPE**	Reframe as the "V2G reality check": measured negligible *under stated assumptions*; needs battery-capacity sensitivity.
VPW (willingness) bidding	**DISCARD** for now	Labels are proxies; no evidence either way. Keep out of main results.
50% cost reduction, 4–10× sample efficiency	**DISCARD**	Not supported; replace by measured, paired, held-out numbers.
Federated learning	**RE-SCOPE**	Retain as a secondary contribution: local EDF tuning + FedProx on the residual trunk; claim only what the 2-site results show (§6).
IEEE-123 coding	**DISCARD** (code) / **KEEP** (theory)	Real IEEE-123 is multi-phase unbalanced; a synthetic 123-bus graph would be criticised. We prove an exact tree decomposition instead (§7).
Optuna on one training day	**DISCARD**	Overfits the date (E10). Any tuning must use the multi-day train/val protocol.

4 Revised contributions and what is provable

Setting. Radial feeder $\mathcal{T}$, LinDistFlow voltages $v = v_0 - 2(RP + XQ)$ and branch flows $S = M(P, Q)$. A rule or a policy proposes a; the safety layer returns $\Pi(a) = \arg\min_x \|x - a\|_W^2$ s.t. $Gx \le h - \tau$ (box, voltage rows, thermal polygon rows), where τ_r is a per-row margin.

Proposition 1 (Row-wise conformal coverage)**.** *Let $e_r = g_r^{\text{AC}} - g_r^{\text{Lin}}$ be the AC-minus-linear constraint residual of row r on n calibration steps exchangeable with a test step, and τ_r = the $\lceil (n+1)(1-\alpha) \rceil$-th smallest of e_r. Then $\Pr[g_r^{\text{AC}} \le h_r \mid \textit{Lin-feasible}] \ge 1 - \alpha$ for each row, and $\ge 1 - m\alpha$ jointly over m rows (Bonferroni).*

Status: standard split-conformal argument; implemented (`ResidualTightening`). **Not** "zero violations guaranteed": it is marginal, per-step, under exchangeability, and degrades under distribution shift — hence the adaptive-conformal extension below. Empirically: 0 AC violations on all held-out days of all runs (E1,E7).

Proposition 2 (Restoration)**.** *Given $x = \Pi_{\text{ADMM}}(a)$ with residual $r = \max(Gx - h)^+$, the shrink $x \leftarrow \theta x$ with the largest $\theta \in [0, 1]$ satisfying $G(\theta x) \le h$ exists whenever $h \ge 0$ (idle feasibility), and the relaxation $h := \max(h, 0)$ is the only change made to rows that were infeasible at idle. Hence the returned action is feasible for the linear model up to float tolerance.*

Proposition 3 (Bounded residual)**.** *Let Π be the weighted-norm projection (non-expansive). If the policy is $a = \text{clip}(b + \beta\, u)$ with $\|u\|_\infty \le 1$, then $\|\Pi(a) - \Pi(b)\|_W \le \beta \|\mathbf{1}\|_W$: the executed action stays within a β-ball of the safe base action. (Observed: cost is monotone in β: +1.8% at 0.1, +6.3% at 0.3.)*

Proposition 4 (Validation gate)**.** *Let $d_i = J_{\text{base}}(day_i) - J_{\text{res}}(day_i) \in [-B, B]$ on n held-out days exchangeable with deployment days. Deploy the residual iff $\bar{d} - B\sqrt{2\ln(1/\delta)/n} > 0$. Then $\Pr[\textit{deploy} \mid \mathbb{E}d \le 0] \le \delta$ (Hoeffding). Hence the deployed controller is, w.p. $\ge 1 - \delta$, not worse than the tuned heuristic in expectation.*

With $n = 4$ test days (today) the gate would never fire — which is the correct, honest outcome of E7 — and $n \ge 30$ is required for it to have power. This is the missing statistics in most RL-for-energy papers (cf. Agarwal et al. 2021).

Contributions

C1. Certified-safe dispatch: row-wise conformal margins + restoration + differentiable QP (Props. 1–2), empirical zero-violation on all held-out days, global-margin failure mode documented.

C2. Benchmark: real ACN days, tariffs, demand charge, stress, multi-day split, oracle and tuned baselines, two sites.

C3. Headroom-gated learning protocol: define $H = (J_{\text{tuned}} - J_{\text{oracle}})/J_{\text{tuned}}$; learn only if H exceeds a pre-registered threshold (we propose 15%); report H for every scenario (here 5–12%: the negative result is itself the contribution and explains why untuned-baseline RL papers over-claim).

C4. Validation-gated safe residual RL: Props. 3–4; honest report of today's 2–9% gap.

C5. Scale-up: exact recursive tree decomposition of the safety layer (§7), with a numerical check of the key lemma.

5 Creative additions (each with a hypothesis, a proof-of-concept and a kill criterion)

A1. Adaptive-conformal margins for non-stationary feeders. Replace the fixed split-conformal quantile by an online update $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$ per row (Gibbs & Candès 2021). *Test:* inject a mid-run load/PV shift; realised violation rate should track α where the fixed margin drifts. *Kill:* no improvement in tracking error.

A2. Per-hub caps as a 3-dimensional action. The three hub branches (buses 16/15/14) carry the binding rows; learn (c_1, c_2, c_3) instead of 108 powers. *Test first:* oracle vs best per-hub-cap heuristic headroom. *Kill:* $< 5\%$.

A3. Forecast-aware MPC baseline (a must-have). Our two online PoCs fail for explicit reasons: day-level context does not choose the cap (E4); price-blind peak shaving ignores the tariff (E5). A rolling-horizon QP with a price/arrival forecast and the same safety layer is the baseline reviewers will ask for (ACN-Sim uses MPC); it also tells us whether any online headroom exists.

A4. QP duals as congestion prices. The KKT multipliers of the safety layer are locational shadow prices; use them as features (or as the EDF budget split across hubs). *Test:* replaces A2's learned caps by dual-priced caps.

A5. Federated gating. Aggregate only if a site's headroom exceeds the threshold; otherwise keep the local heuristic. Federated claims are then conditional and falsifiable.

A6. Paired, day-blocked statistics. All comparisons are per-day paired, bootstrap CIs over days, IQM over seeds (Agarwal et al. 2021). Day-to-day cost std (∼135) dwarfs the policy differences; unpaired tables are not acceptable.

A7. Cost–served–peak Pareto fronts instead of one scalar cost, because the unmet penalty weight (2 \$/kWh) is a modelling choice.

A8. Price-shift robustness: train on one tariff, evaluate on another (SCE stand-in → CAISO-like) to separate tariff over-fitting from site over-fitting.

6 Experimental protocol and results so far

Protocol. Per site: 20 evenly spaced weekdays, seeded 12/4/4 train/val/test split; EDF cap tuned on *local train days only*; checkpoint chosen on val; report on test days; stress = 2× charger power and 2/3 of nameplate hub capacity (Caltech 100 kW, JPL 130 kW); demand charge 1.5 \$/kW-day; dense reward. Tariffs are per-site settings (§8).

Cross-site transfer (train Caltech → test JPL), and Caltech+JPL federated. *(Run: 60 central episodes on Caltech train days, seed 0, residual scale* 0.1 *around the Caltech-tuned cap; JPL evaluated zero-shot on its 8 val+test days, none seen in training. Costs in \$/day, mean over days; "viol" of raw rules is the AC thermal overload in p.u.; evidence* `docs/study/transfer_*.json`*.)*

Regime	Policy	Caltech test	JPL (SCE stand-in)	JPL (CAISO-like synth.)	JPL served
normal	safe-uncontrolled (QP only)	332.9	858.7	732.2	85.5%
	EDF cap, Caltech-tuned (0.45)	297.1	1155.4	1045.1	75.2%
	EDF cap, JPL-tuned (1.0, raw)	—	823.8	694.1	86.5%
	residual SafeTD3 (last)	299.7	1155.0	1044.7	74.9%
	oracle LP	276.6	807.5	674.8	84.9%
stress	safe-uncontrolled (QP only)	313.3	769.9	634.8	91.4%
	EDF cap, Caltech-tuned (0.6)	266.1	1114.7	997.4	80.1%
	EDF cap, JPL-tuned (1.0, raw)	—	795.4	650.0	91.2%
	residual SafeTD3 (last)	270.9	1048.0	924.4	82.2%
	oracle LP	246.1	800.1	661.6	87.8%

What it says (honest reading).

(a) *In-domain* (Caltech test) the residual learner is within 0.9% (normal) and 1.8% (stress) of the tuned EDF — close, not better.

(b) *A tuned heuristic does not transfer.* The best cap is 0.45–0.6 at Caltech but 1.0 (no cap) at JPL (195 kW capacity, different session mix): the Caltech cap costs JPL +35 to +45% versus plain charging through the safety layer.

(c) *The residual inherits its base's failure.* Because the executed action is within a β-ball of the base (Prop. 3), a mis-specified base is a ceiling on robustness. The residual does improve on its own bad base zero-shot (-6 to -7% in the stressed regime, 0% in the normal one) but remains far from the tuning-free baseline.

(d) *The tuning-free QP wrapper is the robust choice:* safe-uncontrolled is within 6% of the oracle at JPL (normal) and *below* the (imperfect, margin-tightened) oracle in the stressed CAISO-like arm; the raw JPL-tuned rule reaches the same cost but violates the feeder (0.03–0.15 p.u.).

(e) *Price shift (SCE stand-in → synthetic CAISO-like) changes levels, not orderings*: the ranking of policies is the same under both tariffs (a weak but useful robustness check; real CAISO data still needed).

Design consequence: the base must be tuned *per site* (locally, no data sharing) and the headroom gate must be evaluated per site: JPL has ≈ 0% headroom over the safety-wrapped plain charger, so nothing should be learned or federated there.

Federated Caltech + JPL (pilot: 8 rounds × 2 local episodes, seed 0). Each client tunes its own EDF cap on its own train days (Caltech 0.45 / 0.6, JPL 1.0: normal / stress) and trains a residual (scale 0.1) around it; FedProx ($\mu = 0.05$) aggregates only the shared trunk. Held-out test days per site (4 each); \$/day:

Regime	Site	local EDF	safe-uncontr.	FedProx	Local-only	Oracle
normal	Caltech	297.1	332.9	299.5	299.2	276.6
normal	JPL	765.8	789.8	776.1	772.3	734.8
stress	Caltech	266.1	313.2	273.2	272.7	246.1
stress	JPL	794.7	687.5	768.6	769.7	684.1

Honest reading. (a) FedProx ≈ local-only to within ±0.5%: *no federation benefit is measurable at this scale*; the pilot has 16 episodes per client and one seed. (b) Both are 0.8–2.7% above the local EDF at Caltech (as in the central runs), and -3% to +1.3% versus the local EDF at JPL; at JPL under stress the (locally tuned) EDF base is itself 16% worse than the tuning-free safe-uncontrolled rule (794.7 vs 687.5), so the right JPL controller is the QP-wrapped plain charger and the gate (Prop. 4) should refuse the residual. (c) The shared trunk drift after FedProx aggregation is 0 by construction (clients are synchronised after each round); local-only drift is ≈ 45, i.e. the trunks do diverge without aggregation, but divergence alone did not change test cost. (d) All runs: zero violations through the safety layer. **Claim we can make today:** federated training is *feasible and safe* across two real, heterogeneous sites with locally tuned bases and without sharing sessions; **claim we cannot make:** that it improves cost. That needs ≥ 5 seeds, ≥ 100 rounds, and a third site.

7 Scale-up: recursive tree decomposition of the safety layer (theory; no 123-bus code)

Why not just code IEEE-123. The real IEEE-123 feeder is multi-phase and unbalanced with regulators and switches; the 123-bus graph in the repository is synthetic and was disabled on purpose. A reviewer would (rightly) reject physics results from it. Moreover, the cost of scaling is not the 123×123 sensitivity matrix (trivial); it is the number of charger variables $2K$ in one dense projection.

Lemma (3-scalar coupling). Cut a radial feeder at branch e_b with subtree S (buses below e_b), boundary bus b. In LinDistFlow:

(i) for every bus $j \notin S$, R_{ij} and X_{ij} are identical for all $i \in S$ (common root paths end above e_b); so the outside world shifts every voltage in S by *one scalar* Δv_b (the boundary voltage);

(ii) for every branch e outside S, M_{ej} is identical for all $j \in S$; so S acts on the rest of the feeder through *one aggregate* injection $(P_S, Q_S) = \sum_{j \in S}(P_j, Q_j)$;

(iii) voltage rows of S are $v_i = v_b - 2\sum_{e \in \mathrm{path}(b \to i)}(r_e P_e + x_e Q_e)$ with P_e, Q_e sums over buses *inside* S; thermal rows of branches inside S are local.

Hence the constraint matrix is block-angular with exactly 3 coupling scalars (v_b, P_S, Q_S) per cut, however large S is. *Numerical check* (`scripts/poc_tree_decomposition.py`): (i)–(ii) hold for 27/27 cut branches of IEEE-33 and 80/80 of a random 123-node tree.

Decomposed projection. Introduce copies $(\hat{v}_b, \hat{P}_S, \hat{Q}_S)$ per cut; consensus ADMM over M blocks gives the *same optimum* as the monolithic QP (convex, exact constraints) with each block solving a QP of size $2K_m$. Conformal margins are row-local, so each block calibrates its own rows; feasibility certificates (Prop. 2) hold per block and compose because coupling is only through the 3 scalars.

Recursion. Apply the cut recursively: a feeder of N buses becomes a tree of subtrees with branching $\leq c$ and depth $O(\log_c N)$; each internal node exchanges 3 scalars with each child. Upward pass (backward sweep) aggregates (P_S, Q_S); downward pass (forward sweep) sends $\hat{v}_b$ — the classical backward/forward structure of radial power flow, now applied to the *projection*. Cost: monolithic dense Cholesky $O((2K)^3)$ vs. $O\big(M(2K/M)^3 \cdot I\big)$ with I consensus iterations; this is $\approx O(K)$ *only if* block sizes are bounded and I is bounded — we do not claim "$O(N)$" without measuring I. *Planned micro-benchmark* (no RL): runtime of monolithic vs. decomposed projection on random radial trees with $N \in \{33, 123, 500, 2000\}$ and K proportional to N; *kill:* no speed-up at $K \geq 500$.

Link to federation. Each facility is a natural block coordinator holding only local sessions; the parent exchanges the 3 boundary scalars, not driver data. Caveat: in our experiments Caltech and JPL are separate feeders; the subtree story is architectural, not evaluated.

8 Datasets: what we need and how the code degrades gracefully

Full specification: `docs/DATASETS_REQUIREMENTS.md`. Summary:

- **Have:** Caltech and JPL ACN-Data CSVs; IEEE-33; SCE TOU-D-PRIME (a *residential* plan at *2026* rates, applied to 2018–2021 days: a stand-in).
- **Need:** (1) the historical commercial tariff and demand-charge structure that Caltech and JPL actually paid (verify each site's serving utility; JPL's need not equal Caltech's); (2) CAISO SP15 DAM/RTM LMP CSV for 2018–2021 (OASIS `PRC_LMP/PRC_INTVL_LMP`, node `TH_SP15_GEN-APND`); (3) a real feeder load profile; (4) a battery-capacity distribution for the V2G what-if.
- **Graceful handling (implemented, tested):** `price_source` is per site/client; `caiso_rtp` reads a real LMP CSV if given (`lmp_csv_price`) and otherwise prints one warning and uses the labelled synthetic duck curve; results are labelled by price source.
- **Assumed (disclose):** battery capacity and arrival SoC, V2G hardware, VPW labels, hub siting, retail adder (0.12 \$/kWh), demand charge value.

9 Anticipated reviewer objections

Objection	Our answer
"The RL baseline is weak / the heuristic is a strawman."	We tune EDF on train days, report the oracle, and *lose* to the heuristic today; the paper's claim does not depend on winning.
"Zero violations is only in your linear model."	Violations are measured against exact AC (BFS) on held-out days; the guarantee is the conformal coverage of Prop. 1, not a blanket claim.
"Two days/seeds are anecdotal."	Multi-day split, 20 days per site, paired CIs; seeds ≥ 5 on GPU; the gate (Prop. 4) needs $n \geq 30$ and we say so.
"Where does V2G value come from?"	It does not, under stated assumptions; we publish the sensitivity sweep over battery capacity, efficiency, degradation and demand charge.
"Synthetic prices / residential tariff on a workplace."	Labelled as such; the real-LMP and historical-tariff slots are implemented and awaiting the files.
"Why not MPC?"	It is baseline A3; ACN-style MPC is the standard comparator and is in the plan.
"Does it scale?"	Exact block-angular lemma + planned micro-benchmark; no unsupported complexity claim.

10 Plan, risks, decisions needed from Prof. Rajesh

Plan (4 weeks). W1: re-run headroom with real LMP/tariffs if obtained; add MPC baseline (A3) and per-hub-cap oracle test (A2). W2: GPU multi-seed (5–10 seeds, 500+ episodes, 40 train days) for residual

SafeTD3 and the gate; adaptive-conformal test (A1). W3: federated/cross-site with gate (A5), tree micro-benchmark. W4: write-up (venues: IEEE TSG, ACM e-Energy, NeurIPS Datasets & Benchmarks — decide after W2 evidence).

Risks. (R1) Headroom stays $< 15\%$ even with real data $\Rightarrow$ the paper is C1+C2+C3 (safety + benchmark + negative result). (R2) Tariff/battery assumptions drive conclusions $\Rightarrow$ sweeps are mandatory. (R3) Two-site federation is thin $\Rightarrow$ add the third ACN site.

Decisions. (1) Accept the re-positioning above? (2) Provide/approve acquisition of historical tariffs and CAISO LMP CSV. (3) Pre-register the headroom threshold (15%?) and the gate δ. (4) Confirm that cross-site/federated runs are now in scope (the earlier instruction was to defer JPL/communication experiments).

A Corrections to the collaborator memos (so the write-up does not inherit them)

- "Conformal calibration guarantees zero violations" $\rightarrow$ marginal, per-step $1-\alpha$ coverage under exchangeability (Prop. 1).
- "Optuna best 291.89 beats safe greedy" $\rightarrow$ training-day validation cost on a different benchmark (E10).
- "CAISO RTP calibrated to 2019 SP15 data" $\rightarrow$ synthetic duck curve; real CAISO CSV not yet obtained.
- "R is block-triangular, no cross-coupling" $\rightarrow$ imprecise; the exact statement is the 3-scalar Lemma (§7).
- "$O(N^3) \rightarrow O(N)$ guaranteed" $\rightarrow$ only for bounded blocks and bounded consensus iterations; to be measured.
- "Guaranteed A*" $\rightarrow$ no such guarantee exists; the claims above are the ones the evidence supports.

B Reproduction

`python sample.py --tests` (68 tests); `scripts/headroom_multiday.py --days 20 --demand 1.5 --stress`; `scripts/run_multiday.sh R1 --residual 0.3 --base edf_tuned`; `scripts/cross_site_fed.py --task {transfer,federated} --regime {normal,stress}`; `scripts/poc_tree_decomposition.py`; `scripts/poc_cap_selection.py`. Evidence JSONs: `docs/study/`. Narrative audit trail: `DOC.md`.

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